Predicting Employee Attrition Using Remote Work Productivity and Workplace Factors

GROUP PROJECT

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1. Updated Problem Statement

Background of the Attrition Problem

Employee attrition remains a critical issue for organizations because high turnover increases recruitment and training costs, reduces team stability, and affects overall productivity. When experienced employees leave, organizations not only lose talent but also institutional knowledge. Therefore, understanding the factors that influence employee attrition is essential for improving retention strategies and organizational performance.

Traditionally, attrition analysis focuses on HR-related factors such as job satisfaction, salary, overtime, work life balance, job level, and years at the company. These variables help explain workplace conditions that may influence an employee's decision to stay or leave.

Motivation for Adding Productivity Variables

With the growth of remote and hybrid work environments, productivity related behaviors have become increasingly important. Metrics such as productivity score, daily work hours, task completion rate, late task ratio, and focus time may provide additional insight into employee engagement and performance patterns. These behavioral factors may complement traditional HR variables and provide a more comprehensive understanding of attrition.

Therefore, incorporating productivity metrics allows the analysis to move beyond static HR attributes and include performance based dimensions.

Integration Approach

The attrition dataset and the productivity dataset originate from different sources and do not share matching employee identifiers. Because of this limitation, direct employee level merging is not possible.

To address this issue, productivity variables were aggregated by experience level. Specifically, average productivity metrics were calculated for each experience group. These aggregated values were then merged with the attrition dataset using TotalWorkingYears as the linking variable.

This approach ensures logical consistency by aligning employees with productivity patterns associated with similar experience levels rather than attempting to match unrelated individuals.

Refined Research Objective

After integrating the datasets, the final modeling table combines traditional HR variables with aggregated productivity metrics.

The refined research objective is: To examine how workplace factors and productivity related variables jointly influence employee attrition, and to determine which factors are most strongly associated with employees leaving the organization.

2. Data Description

2.1 Employee Attrition Dataset

The Employee Attrition dataset was obtained from Kaggle: Employee Attrition and Factors – TheDevastator https://www.kaggle.com/datasets/thedevastator/employee-attrition-and-factors

This dataset contains 1,470 observations and 35 features, representing individual employee level information. The target variable is Attrition, which indicates whether an employee left the company (1 = Yes, 0 = No). This variable is used as the dependent variable in our classification model. Important variables used in this analysis include:

- MonthlyIncome
- JobLevel
- JobSatisfaction
- WorkLifeBalance
- OverTime
- YearsAtCompany
- TotalWorkingYears
- EnvironmentSatisfaction
- MaritalStatus
- Department

These variables represent traditional HR and workplace-related factors that may influence employee turnover.

2.2 Remote Worker Productivity Dataset

The Remote Worker Productivity dataset was also obtained from Kaggle: Remote Worker Productivity Dataset – Ziya07 https://www.kaggle.com/datasets/ziya07/remote-worker-productivity-dataset

This dataset contains multiple observations representing remote workers' performance and behavioral metrics. Key productivity variables include:

- productivity_score
- stress_level
- average_daily_work_hours

- task_completion_rate
- late_task_ratio
- focus_time_minutes
- experience_years

These variables capture behavioral and performance-based dimensions that may be associated with employee engagement and retention. The productivity dataset does not share employee identifiers with the attrition dataset. Therefore, productivity metrics were aggregated by experience level before integration.

3. Data Preprocessing

3.1 Data Cleaning

Before modeling, several data cleaning steps were performed to ensure data quality and consistency.

First, the dataset was checked for missing values across all variables. After merging the datasets and applying the aggregation approach, the final modeling table did not contain missing values. This ensured that the dataset was complete and ready for analysis.

Second, duplicate records were examined. Any duplicate rows identified in the merged dataset were removed to ensure that each employee was represented only once.

Third, data types were reviewed and corrected where necessary. Categorical variables such as Attrition, Department, JobRole, OverTime, Gender, and MaritalStatus were converted to categorical data types to ensure proper handling during modeling. Numerical variables were verified to remain in appropriate numeric formats.

Next, value ranges were inspected to confirm logical consistency. For example, variables such as JobLevel, WorkLifeBalance, and JobSatisfaction were checked to ensure they fell within expected categorical scales. Salary and experience-related variables were also reviewed to confirm reasonable ranges.

Finally, potential outliers were examined, particularly for MonthlyIncome. Extreme values were capped using percentile-based trimming to reduce the influence of unusually high or low income values on the model. This step helped improve model stability without removing valid observations.

3.2 Data Integration

The data integration process is a critical component of this project because the two datasets originate from different sources and do not share a common employee identifier.

The Employee Attrition dataset contains individual employee level HR information, while the Remote Worker Productivity dataset contains performance related records for a separate population of workers. Since there is no shared employee ID between the two datasets, a direct row level merge is not possible. Attempting to join the datasets using employee identifiers would result in incorrect matches and invalid analysis.

To address this limitation, productivity metrics were aggregated by experience level. Specifically, the productivity dataset was grouped by experience_years, and the following average metrics were calculated:

- avg_productivity
- avg_daily_hours
- avg_task_completion
- avg_late_ratio
- avg_focus_minutes

This aggregation created one summarized productivity profile for each experience level. Instead of matching individual employees, the integration aligns employees with the average productivity behavior of workers who have similar years of experience.

The aggregated productivity table was then merged with the attrition dataset using:

TotalWorkingYears ↔ experience_years

A LEFT JOI**N** was used during the merge process. This ensured that all employees from the attrition dataset were retained, even if a corresponding experience level was not present in the productivity dataset.

The number of rows remained consistent before and after the merge, confirming that no employee records were lost during integration. The merge only increased the number of columns by adding aggregated productivity features.

For any unmatched experience levels, missing productivity values were handled by filling them with overall average productivity metrics. This ensured that the final integrated modeling table contained no missing values.

As a result, the final dataset represents an integrated modeling table that combines traditional HR variables with experience level productivity characteristics, allowing for a more comprehensive analysis of employee attrition.

3.3 Data Normalization and Feature Engineering

After data integration and cleaning, several preprocessing steps were applied to prepare the dataset for analysis.

First, the Attrition variable was converted into binary format, where 1 represents employees who left the company and 0 represents employees who stayed. This allowed the variable to be used directly in classification modeling.

Next, categorical variables such as Department, JobRole, OverTime, Gender, MaritalStatus, and EducationField were converted to categorical data types and encoded for modeling purposes. This ensured that non-numeric variables could be properly interpreted by the machine learning algorithm.

Numeric features were standardized using StandardScaler. Since variables such as MonthlyIncome, YearsAtCompany, and the aggregated productivity metrics have different scales, normalization was applied to place them on a comparable range. This step prevents larger scale variables from dominating the model.

Moreover, experience level groups were created from TotalWorkingYears for visualization purposes. These grouped categories helped analyze attrition trends across different career stages, while the original numeric variable was retained for modeling.

4. Preliminary Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure of the data and identify relationships between variables and employee attrition.

4.1 Descriptive Statistics

Summary statistics were used to analyze the distribution of key variables such as income, job satisfaction, and tenure. The analysis indicated that employees who left the organization often had lower job satisfaction and shorter tenure compared to those who stayed.

4.2 Data Visualizations

Several visualizations were used to explore patterns within the data:

4.2.1 Pair Plots

Pair plots were used to examine relationships between productivity metrics, income, and attrition. These visualizations revealed potential differences in productivity and compensation patterns between employees who left and those who remained.

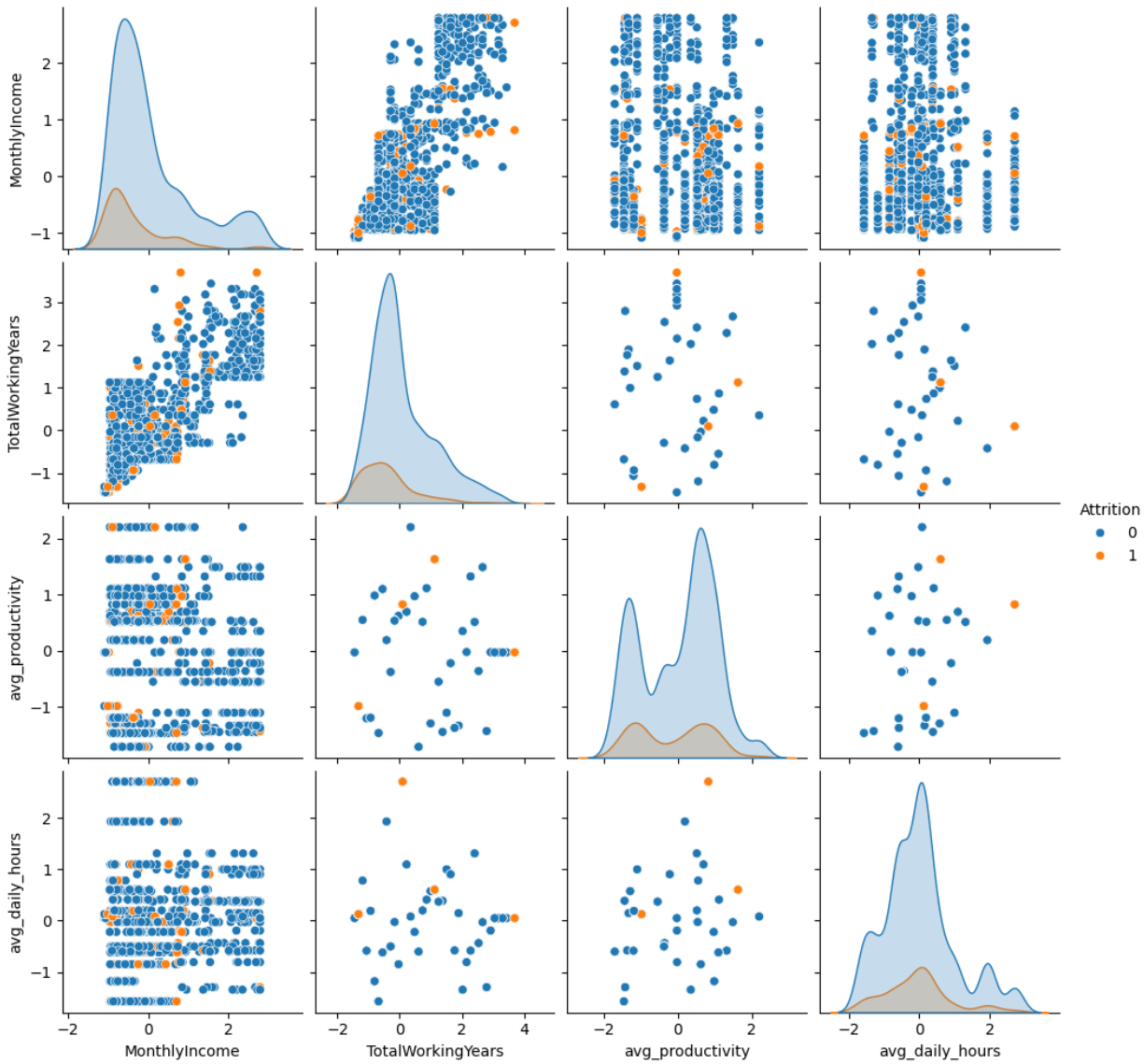

Figure 1. Pairwise Relationships Between Key Variables and Employee Attrition

4.2.2 Boxplots

Boxplots were created to compare distributions of variables across attrition categories. Key observations included:

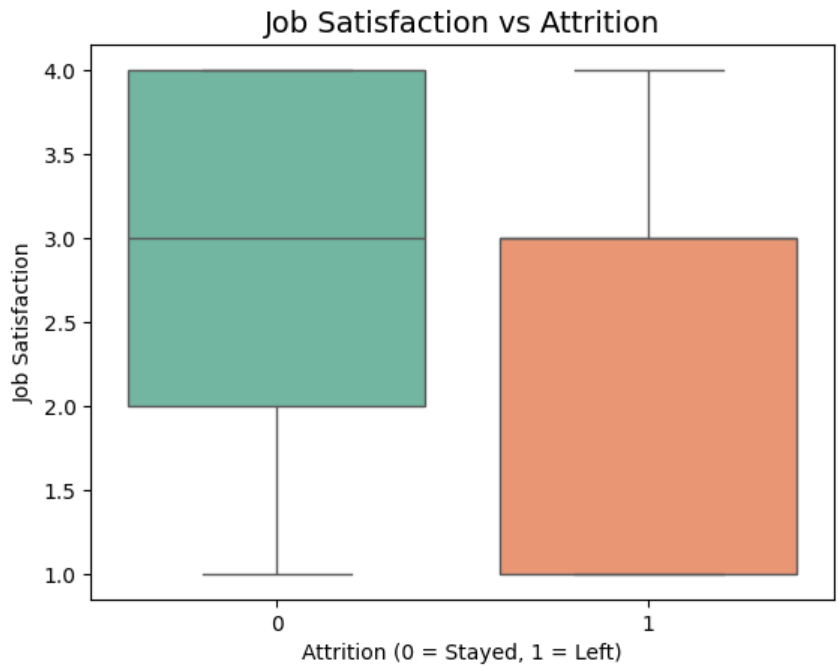

Figure 2. Job Satisfaction by Employee Attrition Status

Employees with lower job satisfaction tended to leave more frequently.

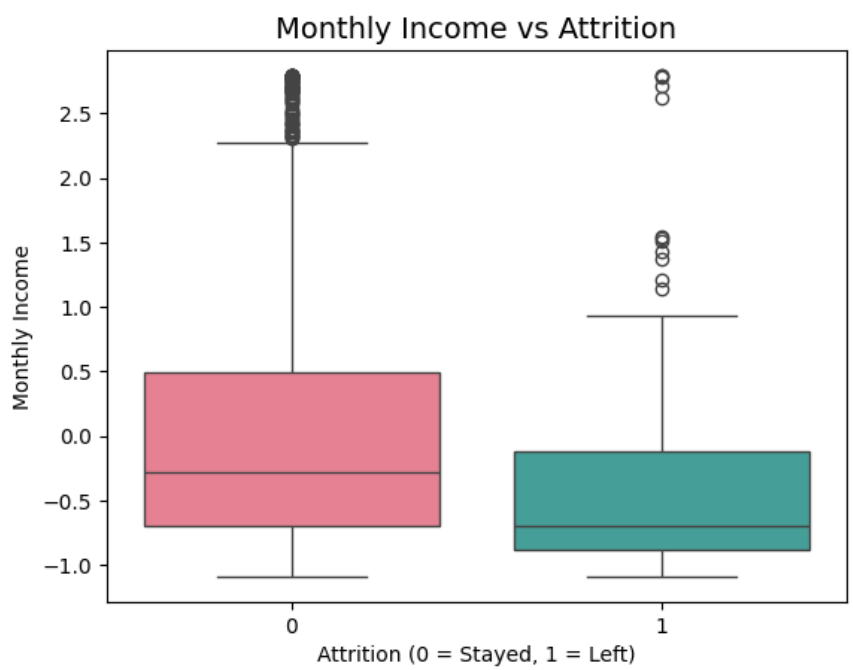

Figure 3. Monthly Income by Employee Attrition Status

Employees with lower monthly income were slightly more likely to leave.

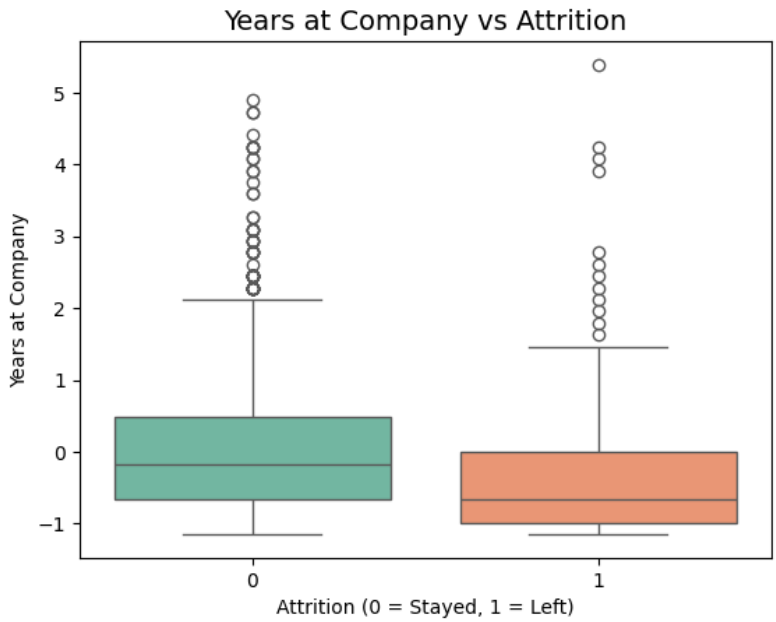

Figure 4. Years at Company by Employee Attrition Status

Employees with fewer years at the company were more likely to exhibit attrition.

4.2.3 Count Plots

Count plots were used to visualize attrition by job level. These plots revealed that employees in lower job levels experienced higher attrition rates compared to those in more senior positions.

Thus, EDA provided valuable insights into potential predictors of attrition and guided feature selection for the predictive models.

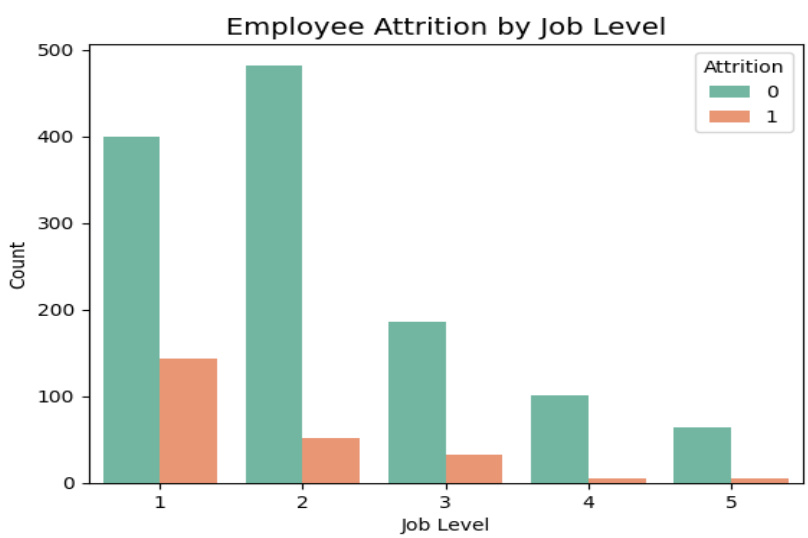

Figure 5. Employee Attrition by Overtime Status

5. Modeling Results

Employee attrition prediction was treated as a binary classification problem, where the goal is to classify employees as either staying or leaving the organization.

Two machine learning algorithms were implemented which were Logistic Regression and Random Forest.

5.1 Modeling Framework

The dataset was divided into training and testing subsets using a standard train-test split. To ensure robust performance estimation, 5-fold cross-validation was also applied during model training.

5.1.1 Evaluation Metrics

Model performance was evaluated using several metrics that includes Accuracy, Precision, Recall, F1-score and ROC-AUC.

Because attrition cases represent a minority class, particular emphasis was placed on precision, recall, and F1-score, which better reflect performance on imbalanced datasets.

5.2 Logistic Regression

Logistic regression was chosen as the baseline model because it provides interpretable coefficients that help identify the influence of individual variables on attrition probability.

The logistic regression model achieved Accuracy: as 88% and ROC-AUC: 0.79

While the model performed well overall, recall for the attrition class was lower, indicating that some employees who leave were not correctly identified.

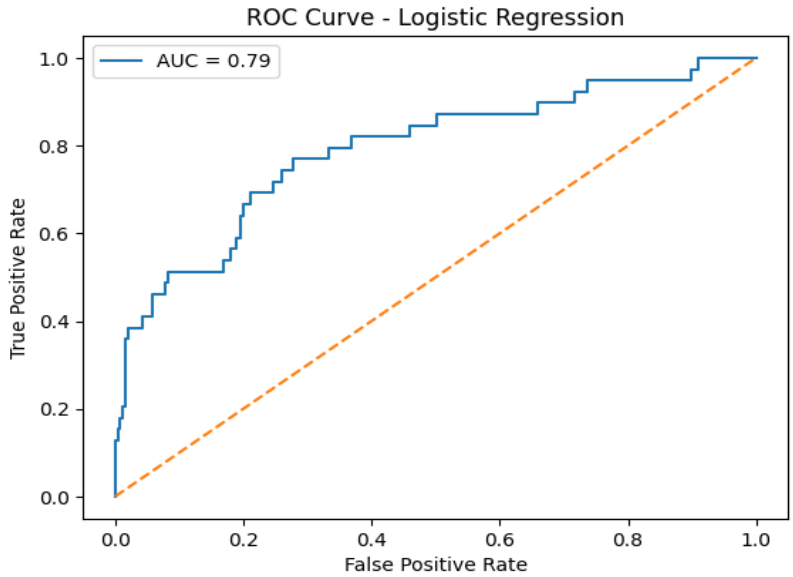

Figure 6. ROC Curve for Logistic Regression Model

```
Logistic Regression Performance Metrics:
Accuracy: 0.881
Precision: 0.577
Recall: 0.385
F1-score: 0.462
ROC-AUC: 0.791
```

5.3 Random Forest

A Random Forest model was implemented to capture nonlinear relationships and interactions between features. Random Forest is an ensemble method that combines multiple decision trees to improve predictive performance.

The Random Forest model achieved ROC-AUC as 0.74 and Accuracy as 88%.

Although Random Forest can capture more complex relationships, its performance in this analysis was slightly lower than logistic regression.

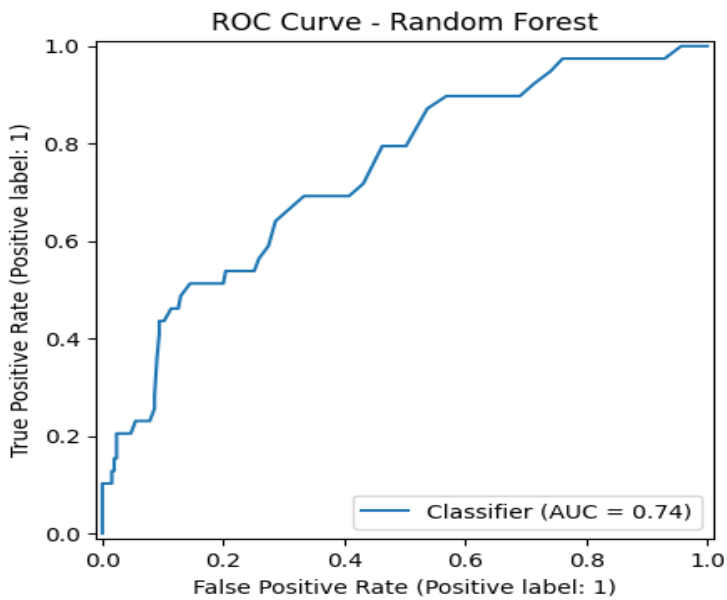

Figure 7. ROC Curve for the Random Forest Model

5.4 Model Comparison

ROC curves were used to compare the classification performance of both models. Logistic regression demonstrated slightly better discrimination between attrition and non-attrition cases.

Additionally, Precision-Recall curves were used to evaluate performance for the minority class. These curves confirmed that logistic regression maintained stronger predictive ability for identifying employees who leave.

5.5 Feature Importance

Feature importance analysis revealed that several variables strongly influence attrition prediction, including:

- Monthly income
- Job satisfaction
- Years at company
- Productivity-related variables

These findings indicate that both workplace conditions and productivity patterns contribute to employee turnover.

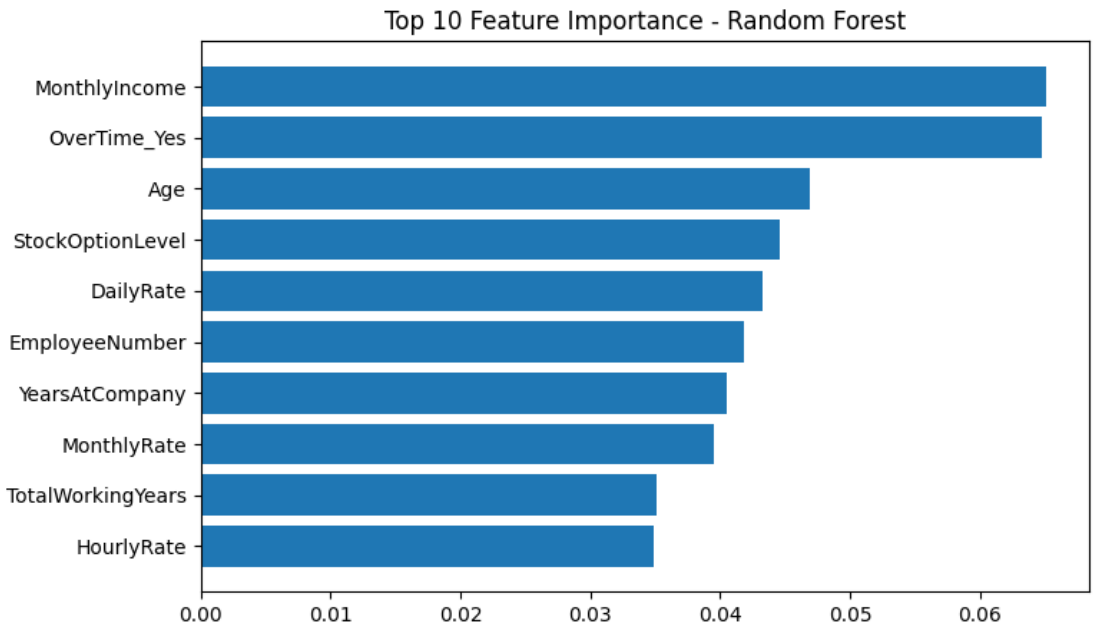

Figure 8. Top 10 Feature Importance from the Random Forest Model

6. Discussion of Results

The experimental results demonstrate that employee attrition can be reasonably predicted using a combination of workplace and productivity-related variables. Logistic regression provided the best balance between predictive performance and interpretability.

Although Random Forest is capable of modeling complex nonlinear relationships, the simpler logistic regression model performed slightly better on this dataset. This suggests that the relationships between the predictors and attrition are relatively linear and well captured by logistic regression.

The analysis also confirmed that traditional HR variables such as income and job satisfaction remain strong predictors of attrition, while productivity and behavioral metrics provide additional context for understanding employee behavior.

Therefore, the results support the initial problem statement by demonstrating that predictive modeling can help identify employees at risk of leaving and provide insights into the factors contributing to turnover.

7. Git Repository

GitHub Repository Link: https://github.com/Kaurgurpreet23/CPSC5071

8. References

8.1 Related work

Previous research shows that employee attrition is influenced by job satisfaction, compensation, workload, and career growth opportunities. Machine learning approaches such as Logistic Regression and Random Forest are commonly used for attrition prediction due to their interpretability and ability to capture nonlinear relationships.

Logistic Regression provides a baseline model for identifying statistically significant predictors of employee turnover, while Random Forest improves predictive performance by modeling complex interactions among workplace factors.

Recent studies also highlight the importance of behavioral and productivity metrics in understanding employee engagement, particularly in remote work environments. Incorporating productivity indicators alongside traditional HR variables allows for a more comprehensive attrition modeling framework.

8.2 Citations

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